Neural Caption Image Retrieval

Two page project write up | 05 October 2026 | Reproduction and extension

Problem and dataset

Given a natural language caption, rank gallery images by how closely their visual content matches its meaning. The project reproduces five methods from Qian and Lamberti's *Neural Caption Image Retrieval* and evaluates a separately trained extension.

MS COCO 2014 uses the Karpathy split: 113,287 training images, 5,000 validation images and 5,000 test images. Five captions per training image provide 566,435 captions. The authors' archive supplies normalized 4,096 dimensional VGG 19 fc7 features from ten crops. Feature order matches the caption manifests. The original vocabulary has 27,011 entries, captions use 50 padded whitespace tokens, and GloVe 6B supplies 300 dimensional initial word vectors.

Original paper methods and training

Two baselines. Average GloVe word vectors and fit a least squares map into VGG feature space. The pooled variant combines five held out captions into one query per image. Both rank images by squared Euclidean distance.

Three recurrent models. GRU, GRU + GloVe and LSTM + GloVe encode captions with the recurrent final state. A linear image projection maps VGG features into the same 1,024 dimensional space. Unit normalized representations train with a bidirectional margin ranking loss of margin 0.1. The PyTorch reproduction follows legacy Keras gate equations and all 50 padded positions.

Each original neural model completed 50 epochs using legacy Adam, learning rate 0.001, batch size 128 and 886 batches per epoch, sampling one caption per image. Validation Recall@10 selected epochs 46, 46 and 38 for the three models. The paper lists 0.05, but the released code does not pass that setting to Adam.

New architecture and training

BiGRU + GloVe. A bidirectional GRU uses 512 units per direction. It pools valid token states by their mean, combines both final states, and projects the concatenation into 1,024 dimensions. A nonlinear image projection uses two layers with normalization, GELU and dropout. Both outputs have unit length. Existing VGG features are reused; the CNN is not retrained.

The extension uses symmetric contrastive loss plus a hardest negative ranking term inspired by VSE++. It trains on the same 113,287 images, with one of five captions sampled per image per epoch, AdamW learning rate 0.0002 and batch size 256. Preset early stopping required at least 20 epochs, allowed at most 50 and used patience eight. Training completed 31 epochs; validation selected epoch 23 with Recall@10 of 87.68%.

Combined model. Concatenating the two normalized encoders with square root weights creates a 2,048 dimensional representation. Original GRU + GloVe epoch 46 receives weight 0.75 and the new BiGRU epoch 23 receives 0.25. Validation alone chose this setting from original weights 0, 0.25, 0.50, 0.75 and 1; validation Recall@10 was 89.64%. Test scores were not used for selection.

Implementation

Python and PyTorch provide reproducible preparation, training, checkpointing and evaluation. The interface defaults to the combined model and offers the original GRU + GloVe. Queries are encoded, compared with gallery representations computed from saved image features on each search, and sorted to display the best matching images.

Evaluation and measured results

Both protocols rank exactly the same 1,000 test images. The main protocol uses one selected caption per image, giving 1,000 queries. The reconstructed released evaluator uses all five captions, giving 5,000 queries. R@1 and R@10 measure the percentage of correct images within the first one and ten results. Lower mean rank is better.

Main protocol: 1,000 image gallery and 1,000 queries

Method	R@1	R@10	Mean rank
Average GloVe baseline	9.5%	45.2%	37.056
Pooled caption baseline*	11.0%	52.2%	25.640
Original GRU	34.5%	82.8%	9.127
Original GRU + GloVe	34.5%	82.6%	9.699
Original LSTM + GloVe	31.7%	83.1%	8.636
New BiGRU + GloVe	36.5%	85.0%	8.666
Combined model	38.6%	85.5%	8.174

* The pooled baseline combines five captions per image; all other rows use one caption. The first five rows reproduce the original methods. The final two rows report the new extension.

Released evaluator: 1,000 image gallery and 5,000 queries

Method	R@1	R@10	Mean rank
Original GRU	36.36%	84.52%	8.5242
Original GRU + GloVe	36.64%	85.04%	8.1462
Original LSTM + GloVe	33.80%	84.96%	7.8108
New BiGRU + GloVe	38.46%	87.34%	7.1098
Combined model	41.20%	88.00%	6.5698

Conclusions and limits

The combined model improves on original GRU + GloVe: main protocol Recall@10 rises from 82.6% to 85.5%, Recall@1 from 34.5% to 38.6%, and mean rank falls from 9.699 to 8.174. With 5,000 queries, Recall@10 rises from 85.04% to 88.00%, a gain of 2.96 percentage points.

The paper reports GRU + GloVe Recall@10 of 86.8%; the combined result of 88.0% is 1.20 percentage points higher. The paper's Table 1 caption selection, original seeds and checkpoints are unavailable, so this is not an exact protocol match or evidence of general superiority. It is a measured improvement over our original implementation on the fixed gallery.

Verification and retrieval example

All 12,000 saved extension ranks were independently recomputed and gallery digests agree. The 1,024 and 2,048 dimensions were verified from actual outputs. All 31 epochs had finite values and 443 batches covering 113,287 samples. Five real interface checks passed for normal text, capitals, punctuation, unknown words and the original model. The combined model places the cyclist beside a train at rank 1 for 'a guy riding a bike next to a train'; 'dog' returns a dog first.

Code: github.com/kishansaaai/ML_PROJECT. Sources: Qian and Lamberti, *Neural Caption Image Retrieval* (CS229); Faghri et al., *VSE++* (2018). Metrics and checks come from the saved original and extension evaluations.